# Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

Russell Ng (A0213913M)
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## 1 Motivation

### 1.1 Origin in fluid dynamics

Burgers' equation was introduced by J. M. Burgers as a simplified model for turbulence [1], isolating the two competing effects that make the Navier–Stokes equations hard to analyse: nonlinear advection, which steepens a velocity profile, and viscous diffusion, which smooths it. The one-dimensional Navier–Stokes momentum equation is

$$u_t + u\,u_x = -\frac{1}{\rho}p_x + \nu\,u_{xx}, \tag{1}$$

where $u(x,t)$ is velocity, $\rho$ density, $p$ pressure, and $\nu > 0$ viscosity. In more than one dimension, $p$ is determined implicitly by the incompressibility constraint $\nabla \cdot u = 0$; this nonlocal coupling is what makes Navier–Stokes hard to control. In one dimension, $u_x = 0$ would force $u$ constant, so the constraint is dropped along with the pressure term entirely, leaving

$$u_t + u\,u_x = \nu\,u_{xx}, \tag{2}$$

the *viscous Burgers' equation*. Setting $\nu = 0$ gives the *inviscid* equation

$$u_t + u\,u_x = 0, \tag{3}$$

which is the main object of study in Sections 2–7, before viscosity is reintroduced in Section 8. What remains after this simplification is a single scalar equation depending only on local values of $u$, but one that retains the nonlinear term $u\,u_x$ responsible for the spontaneous formation of discontinuities studied from Section 3 onward.

*Remark* 1.1 (Vorticity-free flow). Without the constraint $\nabla \cdot u = 0$, the flow may be taken irrotational, so $u = \varphi_x$ for a scalar potential $\varphi$. This is what allows the Cole–Hopf transformation (Section 8) to linearise (2) into the heat equation.

### 1.2 Burgers' equation as a scalar conservation law

Equations (3) and (2) are instances of a *scalar conservation law*,

$$u_t + f(u)_x = 0, \tag{4}$$

with flux $f$, together with its viscous regularisation $u_t + f(u)_x = \nu\,u_{xx}$. For Burgers' equation,

$$f(u) = \frac{u^2}{2}, \tag{5}$$

so $f(u)_x = u\,u_x$ recovers (3). Such equations describe a locally conserved quantity transported at state-dependent speed $f'(u)$, and arise throughout gas dynamics and multiphase flow. Burgers' equation is the simplest nontrivial case (a linear $f$ gives only the trivial transport equation $u_t + c\,u_x = 0$), and is the standard first example for the general theory of characteristics, shocks, weak solutions, and entropy conditions developed in Sections 2–6.

### 1.3 Why study Burgers' equation

Three features make Burgers' equation worth studying in its own right. First, it is the simplest equation whose smooth solutions can lose smoothness in finite time despite smooth data and no forcing; because it is scalar and one-dimensional, this can be analysed completely, unlike the analogous (and in three dimensions still open) questions for Euler and Navier–Stokes. Second, once weak solutions are admitted, it exhibits genuine non-uniqueness (Section 5), requiring an entropy condition (Section 6) to restore well-posedness — a phenomenon common to conservation laws in general, studied here with minimal technical overhead. Third, the viscous equation (2) is one of the few nonlinear PDEs solvable in closed form, via the Cole–Hopf transformation (Section 8), allowing a direct, rigorous comparison between the viscous and inviscid theories.

## 2 Method of Characteristics

This section derives an implicit solution formula for the inviscid Burgers' equation via the method of characteristics. The formula's eventual breakdown is what forces the more general framework of Sections 3–6.

### 2.1 Setup

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{6}$$

with $u_0 \in C^1(\mathbb{R})$. The method of characteristics looks for curves $x(t)$ along which the PDE reduces to an ODE for $u(x(t),t)$.

### 2.2 Derivation of the characteristic equations

For a solution $u$ of (6) and a curve $x(t)$, the chain rule gives

$$\frac{d}{dt}\,u(x(t),t) = u_x(x(t),t)\,\dot{x}(t) + u_t(x(t),t). \tag{7}$$

Since $u_t = -u\,u_x$ on solutions, choosing $x(t)$ to satisfy

$$\dot{x}(t) = u(x(t),t) \tag{8}$$

reduces (7) to

$$\frac{d}{dt}\,u(x(t),t) = 0. \tag{9}$$

Curves satisfying (8) are *characteristics*, and $u$ is constant along each one.

### 2.3 Characteristics are straight lines

**Proposition 2.1** (Characteristics are straight lines)**.** *Let $u$ be a $C^1$ solution of (6) on $\mathbb{R}\times[0,T)$, and let $x(t)$ solve (8) with $x(0)=x_0$. Then for $0 \le t < T$,*

$$u(x(t),t) = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)\,t. \tag{10}$$

*Proof.* By (9), $u(x(t),t) \equiv u(x_0,0) = u_0(x_0)$. Substituting this constant into (8) gives $\dot{x}(t) = u_0(x_0)$, which integrates to the stated formula for $x(t)$. □

Thus any classical solution is completely determined by (10), reducing its construction to inverting $x_0 \mapsto x = x_0 + u_0(x_0)\,t$ at each fixed $t$.

## 2.4 The implicit solution formula

**Theorem 2.2** (Implicit solution via characteristics)**.** *Let $u_0 \in C^1(\mathbb{R})$. For each $t \geq 0$ such that $\Phi_t : \mathbb{R} \to \mathbb{R}$, $\Phi_t(x_0) = x_0 + u_0(x_0)\,t$, is a bijection with $C^1$ inverse, the function $u$ defined implicitly by*

$$u(x,t) = u_0(x_0), \qquad \text{where } x = x_0 + u_0(x_0)\,t, \tag{11}$$

*is the unique $C^1$ solution of (6) at time $t$.*

*Proof. Well-definedness.* Bijectivity of $\Phi_t$ means $u(x,t) = u_0(\Phi_t^{-1}(x))$ is well defined, and $C^1$ since $\Phi_t^{-1}$ and $u_0$ are.

*The PDE holds.* Write $x_0 = x_0(x,t) = \Phi_t^{-1}(x)$. Differentiating $x = x_0 + u_0(x_0)\,t$ in $x$ and in $t$ gives

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\,t}, \qquad \frac{\partial x_0}{\partial t} = \frac{-u_0(x_0)}{1 + u_0'(x_0)\,t}, \tag{12}$$

both well defined since $\Phi_t'(x_0) = 1 + u_0'(x_0)t \neq 0$. By the chain rule applied to $u(x,t) = u_0(x_0(x,t))$,

$$u_x = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad u_t = \frac{-u_0(x_0)\,u_0'(x_0)}{1 + u_0'(x_0)\,t}, \tag{13}$$

so $u_t + u\,u_x = 0$, and $u(x,0) = u_0(x)$ since $\Phi_0 = \mathrm{id}$.

*Uniqueness.* Any $C^1$ solution must, by Proposition 2.1, agree with (11) along every characteristic. □

**Corollary 2.3** (Local existence)**.** *For $u_0 \in C^1(\mathbb{R})$ with $u_0'$ bounded below, there exists $T > 0$ such that (6) has a unique $C^1$ solution on $\mathbb{R} \times [0,T)$, given by (11).*

*Proof sketch.* It suffices that $\Phi_t'(x_0) = 1 + u_0'(x_0)t$ stay bounded away from 0 uniformly in $x_0$ for $t \in [0,T)$; monotonicity of $\Phi_t$ then gives bijectivity with $C^1$ inverse. Let $m = \inf_{x_0} u_0'(x_0)$. If $m \geq 0$, then $\Phi_t'(x_0) \geq 1$ for all $t \geq 0$ and $T = \infty$ works. If $m < 0$, then $\Phi_t'(x_0) \geq 1 + mt > 0$ for $t < -1/m$, so any $T \leq -1/m$ works. □

The bound $T \leq -1/m$ is not an artefact of the proof: it is the exact breakdown time, as Proposition 3.1 shows.

## 2.5 Worked example

Let $u_0(x) = -\arctan(x)$, so $u_0'(x) = -1/(1+x^2) \in [-1, 0)$. By Theorem 2.2, the solution satisfies

$$u(x,t) = -\arctan(x_0), \qquad x = x_0 - \arctan(x_0)\,t, \tag{14}$$

valid while $\Phi_t'(x_0) = 1 - t/(1+x_0^2) > 0$ for all $x_0$. Since $1/(1+x_0^2)$ is maximised at $x_0 = 0$, this holds exactly for $t < 1$: the characteristic through the origin, carrying the steepest slope $u_0'(0) = -1$, is the first to cause $\Phi_t$ to fail to be injective, at $t = 1$.

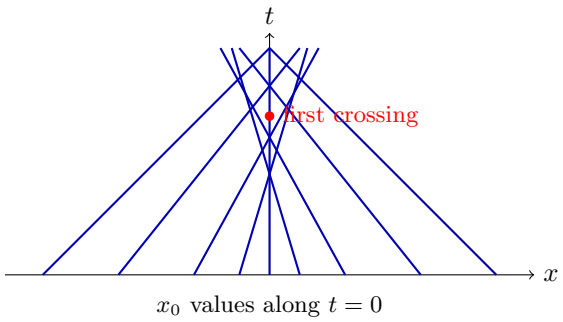



Figure 1: Characteristics for decreasing initial data $u_0$. Where $u_0$ is decreasing, characteristics converge; the first crossing time is $T = -1/\min u_0'$ (Corollary 2.3), computed precisely in Section 3.

*Remark* 2.4 (What happens beyond $T$). Once two characteristics carrying different values of $u_0$ meet, formula (11) assigns $u$ two values at once, so no single-valued classical solution can exist beyond this time. Making sense of the equation past this point is the subject of Sections 3 and 4.

# 3 Shock Formation

This section computes the exact time and location at which the classical solution of Section 2 first breaks down.

## 3.1 Blow-up of the derivative

From the proof of Theorem 2.2,

$$u_x(x,t) = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad x = x_0 + u_0(x_0)\,t. \tag{15}$$

If $u_0'(x_0) \geq 0$, the denominator stays $\geq 1$ and $u_x$ remains bounded along that characteristic for all $t \geq 0$. If $u_0'(x_0) < 0$, the denominator vanishes at the finite time $t = -1/u_0'(x_0)$, where (15) shows $|u_x| \to \infty$: the solution value stays bounded, but its slope becomes vertical.

## 3.2 The envelope of the characteristics

The characteristics form a one-parameter family of lines

$$x(x_0,t) = x_0 + u_0(x_0)\,t, \qquad x_0 \in \mathbb{R}. \tag{16}$$

The first global breakdown occurs where neighbouring characteristics meet soonest, i.e. on the envelope of this family, found by solving (16) together with $\partial x/\partial x_0 = 0$:

$$1 + u_0'(x_0)\,t = 0. \tag{17}$$

This is exactly the vanishing of $\Phi_t'(x_0)$, so the envelope condition coincides with the derivative blow-up condition above.

## 3.3 The shock formation time

**Proposition 3.1** (Shock formation time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \min_{x_0} u_0'(x_0) < 0$, attained at $x_0 = a$. Then the classical solution exists and is $C^1$ on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{18}$$

*and fails to extend as a single-valued $C^1$ function beyond $T^*$. The first breakdown occurs at*

$$x^* = a + u_0(a)\,T^*, \qquad t^* = T^*. \tag{19}$$

*Proof.* Since $s \mapsto -1/s$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every $x_0$ with $u_0'(x_0) < 0$, with equality only at $x_0 = a$. So $T^*$ is the smallest time at which any characteristic meets a neighbour: $\Phi_t'(x_0) > 0$ for all $x_0$ throughout $[0, T^*)$, giving a classical solution there by Theorem 2.2, while $\Phi_{T^*}'(a) = 0$ makes $\Phi_{T^*}$ fail to be locally invertible at $a$, so $u_x$ is unbounded near $(x^*, t^*)$ by (15). Substituting $x_0 = a$, $t = T^*$ into (16) gives $x^*$. □

## 3.4 Worked example, continued

For $u_0(x) = -\arctan(x)$, $u_0'(x_0) = -1/(1 + x_0^2)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$, matching Section 2.5. The shock location is

$$x^* = 0 + (-\arctan(0)) \cdot 1 = 0, \tag{20}$$

so the singularity first appears at $(x^*, t^*) = (0, 1)$, matching Figure 1.

## 3.5 From breakdown to weak solutions

Proposition 3.1 shows the classical formulation of (6) cannot describe decreasing initial data beyond $T^*$, even though nothing in the underlying physics suggests the evolution should stop. Section 4 resolves this by weakening the notion of solution: rather than requiring $u_t + u\,u_x = 0$ pointwise, $u$ need only satisfy an integral identity against smooth test functions, which admits discontinuous solutions for all $t \geq 0$.

# 4 Weak Solutions and the Integral Form

Since classical solutions of (6) cannot be extended past the shock time $T^*$ of Proposition 3.1, this section introduces a weaker notion of solution that admits discontinuities, and derives the condition such discontinuities must satisfy.

## 4.1 The integral formulation

Let $u$ be a $C^1$ solution of $u_t + f(u)_x = 0$ on $\mathbb{R} \times (0,\infty)$ with $f(u) = u^2/2$, and let $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ be any test function, compactly supported but not required to vanish at $t = 0$. Multiplying the PDE by $\varphi$, integrating over $\mathbb{R} \times (0,\infty)$, and integrating by parts in both $x$ and $t$ (using that $\varphi$ has compact support, so all boundary terms at spatial infinity vanish) gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx\, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx, \tag{21}$$

where the boundary term at $t = 0$ survives because $\varphi(x,0)$ need not vanish. Equation (21) makes sense whenever $u \in L^\infty$, with no differentiability required of $u$ at all: this motivates the following definition.

**Definition 4.1** (Weak solution)**.** A function $u \in L^\infty(\mathbb{R} \times (0,\infty))$ is a *weak solution* of (6) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx = 0 \tag{22}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$.

## 4.2 Consistency with classical solutions

**Proposition 4.2** (Consistency)**.** *If $u \in C^1(\mathbb{R} \times [0,\infty))$ satisfies* (6) *classically, it is a weak solution. Conversely, if a weak solution $u$ happens to lie in $C^1(\mathbb{R} \times [0,\infty))$, it satisfies* (6) *classically.*

*Proof sketch.* The first claim is precisely the computation in (21), read in reverse. For the converse, reversing the integration by parts shows a $C^1$ weak solution satisfies $\int (u_t + f(u)_x)\varphi + \int (u(x,0) - u_0(x))\varphi(x,0) = 0$ for every $\varphi \in C_c^1$; taking $\varphi$ supported away from $t = 0$ forces $u_t + f(u)_x = 0$ pointwise by the fundamental lemma of the calculus of variations, and taking $\varphi(\cdot,0)$ arbitrary then forces $u(x,0) = u_0(x)$. □

## 4.3 The Rankine–Hugoniot condition

The purpose of Definition 4.1 is to admit solutions with jump discontinuities. The next result identifies exactly what constraint such a jump must satisfy.

**Theorem 4.3** (Rankine–Hugoniot condition)**.** *Let $s : [0,\infty) \to \mathbb{R}$ be $C^1$, and suppose $u$ is a weak solution of* (6) *that is $C^1$ on each side of the curve $x = s(t)$, with one-sided limits*

$$u_\ell(t) = \lim_{x \to s(t)^-} u(x,t), \qquad u_r(t) = \lim_{x \to s(t)^+} u(x,t) \tag{23}$$

*existing and $u_\ell(t) \neq u_r(t)$. Then $u$ satisfies $u_t + f(u)_x = 0$ classically away from the curve, and along the curve*

$$s'(t) = \frac{f(u_\ell(t)) - f(u_r(t))}{u_\ell(t) - u_r(t)}. \tag{24}$$

*Proof.* Fix $\varphi \in C_c^1(\mathbb{R} \times (0,\infty))$ supported in a region crossed by the curve $x = s(t)$, and split the domain of integration in (22) into $\Omega_\ell = \{x < s(t)\}$ and $\Omega_r = \{x > s(t)\}$, on each of which $u$ is $C^1$. On each piece, integrate by parts in the reverse direction to (21), which is valid since $u$ is $C^1$ there; this produces the interior term $\iint (u_t + f(u)_x)\varphi$ (which vanishes if $u$ solves the PDE classically away from the curve, established by the same interior argument as in Proposition 4.2 applied to test functions supported in $\Omega_\ell$ or $\Omega_r$ alone) plus a boundary term along $x = s(t)$ from each side. Since the outward unit normal to $\Omega_\ell$ along the curve is proportional to $(1, -s'(t))$ (and to $(-1, s'(t))$ for $\Omega_r$), these boundary contributions combine to give

$$\int_0^\infty \Big[\big(f(u_\ell(t)) - f(u_r(t))\big) - s'(t)\big(u_\ell(t) - u_r(t)\big)\Big]\varphi(s(t),t)\, dt = 0. \tag{25}$$

Since $\varphi(s(t),t)$ is an arbitrary $C^1$ function of $t$ along the curve, the bracketed term must vanish identically, which is (24). □

For Burgers' flux $f(u) = u^2/2$, (24) simplifies using $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$ to the familiar average-speed formula

$$s'(t) = \frac{u_\ell(t) + u_r(t)}{2}. \tag{26}$$

*Remark* 4.4. Equation (26) already appears implicitly in Section 3: at the moment of first breakdown, the emerging discontinuity moves at the average of the characteristic speeds $u_\ell$ and $u_r$ arriving from either side, consistent with the characteristic picture of Figure 1. Section 7 uses (26) to construct explicit shock solutions.

# 5 Non-Uniqueness of Weak Solutions

Definition 4.1 was introduced to admit discontinuous solutions past the shock time of Proposition 3.1. This section shows that it admits *too many*: a single initial condition can have more than one weak solution, so the weak formulation alone does not determine the evolution.

## 5.1 The Riemann problem

Consider (6) with the specific piecewise-constant data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{27}$$

This is the simplest instance of a *Riemann problem*, and is studied in general in Section 7. Since $u_0$ is increasing across $x = 0$, the method of characteristics of Section 2 does not directly apply: characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so no characteristic covers the region $0 < x < t$ at all. Two different weak solutions are constructed below, both consistent with this characteristic picture away from $0 < x < t$.

## 5.2 A shock solution

Define

$$u_A(x,t) = \begin{cases} 0, & x < st, \\ 1, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2} = \frac{0+1}{2} = \frac{1}{2}, \tag{28}$$

where $s$ is chosen exactly to satisfy the Rankine–Hugoniot condition (26) with $u_\ell = 0$, $u_r = 1$.

**Proposition 5.1.** *$u_A$ is a weak solution of* (6) *with data* (27).

*Proof.* Away from the line $x = st$, $u_A$ is locally constant, hence $C^1$ and trivially satisfies $u_t + f(u)_x = 0$. Along $x = st$, $u_A$ has a jump from $u_\ell = 0$ to $u_r = 1$ moving at exactly the Rankine–Hugoniot speed $s$. By Theorem 4.3 (read in the direction verified in its proof: a piecewise $C^1$ function solving the PDE on each side and satisfying (24) across the jump satisfies the weak formulation (22)), $u_A$ is a weak solution. □

## 5.3 A rarefaction solution

Define instead

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t. \end{cases} \tag{29}$$

**Proposition 5.2.** *$u_B$ is a weak solution of* (6) *with data* (27).

*Proof.* On each of the three open regions $x < 0$, $0 < x < t$, and $x > t$, $u_B$ is $C^1$. On $0 < x < t$, direct differentiation gives $(u_B)_t = -x/t^2$ and $(u_B)_x = 1/t$, so

$$(u_B)_t + u_B(u_B)_x = -\frac{x}{t^2} + \frac{x}{t} \cdot \frac{1}{t} = 0, \tag{30}$$

and the PDE holds classically on each piece. Moreover $u_B$ is *continuous* across both boundary lines $x = 0$ and $x = t$ (both one- sided limits equal 0 at $x = 0$ and 1 at $x = t$), unlike $u_A$. Repeating the boundary computation in the proof of Theorem 4.3 across each of these lines, the jump terms $f(u_\ell) - f(u_r) - s'(u_\ell - u_r)$ vanish identically because $u_\ell = u_r$ there (there is no jump to begin with, regardless of the line's speed $s'$). Hence all boundary contributions vanish and $u_B$ satisfies (22). □

### 5.4 Two solutions, one initial condition

**Corollary 5.3** (Non-uniqueness)**.** *The initial value problem* (6) *with data* (27) *has at least two distinct weak solutions,* $u_A$ *and* $u_B$*.*

This is a genuine failure of uniqueness, not merely a technicality: $u_A$ and $u_B$ differ on the entire wedge $0 < x < t$, and both are constructed from, and consistent with, the same tools developed so far — $u_A$ from the Rankine–Hugoniot condition of Section 4, and $u_B$ from a direct (if unmotivated) guess later justified by the general construction of Section 7. Definition 4.1 alone gives no way to discard either one.

*Remark* 5.4. Physically, only $u_B$ is correct: for the fluid interpretation of Section 1, increasing data $u_\ell < u_r$ should disperse smoothly (a rarefaction), not jump discontinuously ($u_A$ is, in effect, a shock running *backwards* in time). Section 6 makes this precise with a condition that $u_A$ fails and $u_B$ satisfies, restoring uniqueness.

## 6 The Entropy Condition

Corollary 5.3 showed that the weak formulation alone does not determine a unique solution. This section introduces an additional condition, motivated by the vanishing-viscosity limit made precise in Section 8, that selects the physically correct weak solution among any competing candidates.

### 6.1 The Lax entropy condition

**Definition 6.1** (Lax entropy condition)**.** Let $u$ be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with $f$ convex, having a jump discontinuity along a $C^1$ curve $x = s(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax entropy condition* if

$$f'(u_\ell(t)) > s'(t) > f'(u_r(t)) \tag{31}$$

for all $t$. A weak solution is an *entropy solution* if every jump discontinuity it contains satisfies (31).

For Burgers' equation, $f'(u) = u$, so (31) together with the Rankine–Hugoniot speed $s' = (u_\ell + u_r)/2$ reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{32}$$

*Remark* 6.2. Condition (31) says the characteristics on either side of the shock must run *into* it as $t$ increases, rather than out of it: information is being destroyed at the shock (consistent with physical irreversibility), not created. This is the precise sense in which a shock differs from a rarefaction, where characteristics instead fan *out* of a single point (Section 5.3).

### 6.2 Application to the Riemann problem of Section 5

**Proposition 6.3.** *For the Riemann data* (27) *(*$u_\ell = 0 < u_r = 1$*), the shock solution* $u_A$ *of* (28) *is not an entropy solution, while the rarefaction solution* $u_B$ *of* (29) *is.*

*Proof.* For $u_A$, condition (32) requires $u_\ell > u_r$, but here $u_\ell = 0 < 1 = u_r$, so (31) fails: $u_A$ is not an entropy solution. The function $u_B$ contains no jump discontinuity at all (Proposition 5.2 shows it is continuous), so Definition 6.1 imposes no condition on it, and it is vacuously an entropy solution. □

Proposition 6.3 recovers, as a formal criterion, the physical judgement made informally at the end of Section 5: among the two weak solutions constructed there, the entropy condition selects exactly the rarefaction $u_B$ and rules out the backwards-running shock $u_A$.

### 6.3 Uniqueness

The entropy condition is not merely a rule of thumb for discarding individual bad solutions: it in fact restores full well-posedness.

**Theorem 6.4** (Oleinik, uniqueness of entropy solutions; see [2], §3.4)**.** *For* $u_0 \in L^\infty(\mathbb{R})$ *and* $f$ *uniformly convex, the initial value problem* $u_t + f(u)_x = 0$*,* $u(\cdot, 0) = u_0$*, has at most one entropy solution satisfying* (31) *at every jump.*

The proof of Theorem 6.4 (via an $L^1$-contraction argument comparing two entropy solutions) is beyond the scope of this paper and is not reproduced here; it is stated to record that the selection principle of Definition 6.1 is exactly strong enough to recover uniqueness, and no stronger condition is needed.

# 7 The Riemann Problem

Sections 5–6 treated the specific data $u_\ell = 0 < u_r = 1$. This section gives the complete entropy solution for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, i.e. the general *Riemann problem*

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{33}$$

## 7.1 The rarefaction fan

The solution $u_B(x,t) = x/t$ constructed in Section 5.3 is a special case of a *self-similar* solution, i.e. one of the form $u(x,t) = g(x/t)$. Writing $\xi = x/t$ and substituting into the PDE gives

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{34}$$

so any self-similar solution satisfies either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$) on each interval of $\xi$. This is exactly the structure needed to connect two constant states, provided $u_\ell < u_r$: since characteristic speed equals $u$, the states $u_\ell$ and $u_r$ correspond to characteristics of speed $u_\ell$ and $u_r$, and the fan $u = x/t$ smoothly interpolates the speeds in between whenever $u_\ell < u_r$.

## 7.2 The complete solution

**Theorem 7.1** (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (6) *with data* (33) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{35}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{36}$$

*Proof. (a)* As in Proposition 5.1, $u$ is locally constant away from $x = st$ and jumps at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ across it, so $u$ is a weak solution by Theorem 4.3. Since $u_\ell > u_r$, the Lax condition (32) holds, so $u$ is an entropy solution; uniqueness is Theorem 6.4.

*(b)* The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. On each of the three regions in (36), $u$ is $C^1$ and solves the PDE classically: on the outer two by inspection, and on the fan $u_\ell t < x < u_r t$ by Section 7.1. At the two interior boundaries $x = u_\ell t$ and $x = u_r t$, $u$ is continuous (both one-sided limits agree, equalling $u_\ell$ and $u_r$ respectively), so exactly as in Proposition 5.2 the boundary terms in the weak formulation vanish and $u$ is a weak solution. It contains no jump discontinuity, so it is vacuously an entropy solution, and uniqueness is again Theorem 6.4. □

Taking $(u_\ell, u_r) = (0, 1)$ in case (b) recovers exactly $u_B$ from Section 5.3, and taking $(u_\ell, u_r) = (1, 0)$ in case (a) gives the shock $s = 1/2$ analogous to $u_A$ — now correctly identified as the entropy solution, since the roles of $u_\ell$ and $u_r$ are reversed from Section 5 and the Lax condition (32) now holds.

# 8 Extensions

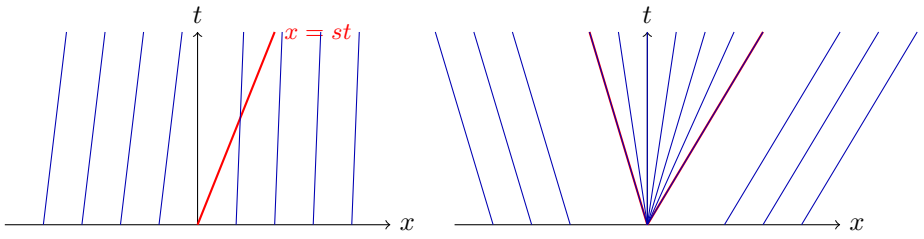
## 8.1 The Cole–Hopf transformation

This extension reintroduces viscosity and solves the viscous Burgers' equation (2) exactly, using the potential-flow structure noted in Remark 1.1.

**Theorem 8.1** (Cole–Hopf transformation)**.** *Let $u$ solve $u_t + u\,u_x = \nu\,u_{xx}$, and define $\varphi > 0$ by*

$$u = -2\nu\,\frac{\varphi_x}{\varphi}, \qquad \textit{i.e.} \qquad \varphi(x,t) = \exp\left(-\frac{1}{2\nu}\int_0^x u(y,t)\,dy\right). \tag{37}$$

*Then $\varphi$ solves the heat equation $\varphi_t = \nu\,\varphi_{xx}$.*




(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristic pictures for the two cases of Theorem 7.1. (a) Characteristics from both sides converge into the shock curve $x = st$. (b) Characteristics fan out from the origin, filling the gap left by the diverging characteristics from $u_\ell$ and $u_r$.

*Proof.* Write $\psi = \ln \varphi$, so $u = -2\nu\,\psi_x$. Substituting into the viscous Burgers' equation and dividing by $-2\nu$ gives

$$\psi_{xt} - 2\nu\,\psi_x\psi_{xx} - \nu\,\psi_{xxx} = 0, \tag{38}$$

which is exactly $\partial_x\big[\psi_t - \nu\,\psi_{xx} - \nu\,(\psi_x)^2\big] = 0$. Hence $\psi_t - \nu\,\psi_{xx} - \nu\,(\psi_x)^2 = C(t)$ for some function $C$ of $t$ alone; since $u = -2\nu\,\psi_x$ is unchanged if $\psi$ is shifted by a function of $t$, $\varphi$ may be rescaled by $\exp\big(-\int_0^t C\big)$ without affecting $u$, so we may take $C \equiv 0$:

$$\psi_t = \nu\,\psi_{xx} + \nu\,(\psi_x)^2. \tag{39}$$

Since $\varphi = e^{\psi}$ gives $\varphi_t = \varphi\,\psi_t$ and $\varphi_{xx} = \varphi\big(\psi_{xx} + (\psi_x)^2\big)$, equation (39) is equivalent to $\varphi_t = \nu\,\varphi_{xx}$. □

## 8.2 Exact solution via the heat kernel

By Theorem 8.1, solving (2) reduces to solving the heat equation for $\varphi$ with initial data

$$\varphi_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right), \tag{40}$$

which is given by convolution with the heat kernel,

$$\varphi(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\varphi_0(y)\,dy. \tag{41}$$

Substituting (40)–(41) into $u = -2\nu\,\varphi_x/\varphi$ and differentiating under the integral gives the explicit *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(s)\,ds + \frac{(x-y)^2}{2t}. \tag{42}$$

**Corollary 8.2** (Global smoothness)**.** *For every bounded measurable $u_0$ and every $\nu > 0$, formula* (42) *defines the unique bounded solution of* (2)*, and this solution is $C^\infty$ for all $t > 0$.*

Unlike the inviscid equation (Proposition 3.1), solutions of (2) stay smooth for all time, for any $\nu > 0$ however small: advection and viscous diffusion together do not by themselves produce singularities.

## 8.3 The vanishing viscosity limit

The Cole–Hopf solution also explains why the Lax entropy condition of Section 6 is the physically correct selection principle, rather than an arbitrary one.

**Theorem 8.3** (Vanishing viscosity limit; Hopf, 1950)**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let $u^\nu$ denote the solution of* (2) *with data $u_0$ given by* (42)*, and let $u$ denote the entropy solution of* (3) *with the same data (Theorem 7.1, extended to general $u_0$ by the same selection principle). Then $u^\nu \to u$ pointwise almost everywhere, and in $L^1_{\mathrm{loc}}$, as $\nu \to 0^+$ [3].*

The proof (a Laplace-method asymptotic analysis of (42) as $\nu \to 0$, showing the integral concentrates near the $y$ minimising $G(x,y,t)$) is not reproduced here. Its significance is conceptual: the entropy condition of Definition 6.1 is not an ad hoc rule imposed to force uniqueness, but exactly the criterion selected by the physically natural limit of vanishing viscosity — the same limit by which the inviscid equation (3) was justified as an idealisation of (2) in Section 1.

## 8.4 A numerical scheme: Godunov's method

This extension outlines a numerical scheme for the conservation law $u_t + f(u)_x = 0$ that uses the exact Riemann solution of Theorem 7.1 as its building block, and states its convergence to the entropy solution.

### 8.4.1 Finite volume discretisation

Divide $\mathbb{R}$ into cells $[x_{j-1/2}, x_{j+1/2}]$ of width $\Delta x$, with cell averages

$$U_j^n \approx \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^n)\, dx, \tag{43}$$

at times $t^n = n\Delta t$. Integrating $u_t + f(u)_x = 0$ over a cell and over $[t^n, t^{n+1}]$ gives the exact identity

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x} \left( \widehat{f}_{j+1/2} - \widehat{f}_{j-1/2} \right), \tag{44}$$

where $\widehat{f}_{j+1/2}$ is the (time-averaged) flux through the interface $x_{j+1/2}$, which must be approximated by a *numerical flux function* $\widehat{f}_{j+1/2} = \widehat{f}(U_j^n, U_{j+1}^n)$.

### 8.4.2 The Godunov flux

Godunov's method defines $\widehat{f}$ by solving, at each interface, the Riemann problem with left and right states $U_j^n$ and $U_{j+1}^n$ *exactly*, using Theorem 7.1, and evaluating the resulting self-similar solution along $x/t = 0$:

$$\widehat{f}_{j+1/2} = f\big(u^R(0;\, U_j^n, U_{j+1}^n)\big), \tag{45}$$

where $u^R(\xi; u_\ell, u_r)$ denotes the value at self-similar coordinate $\xi = x/t$ of the entropy solution from Theorem 7.1 with data $(u_\ell, u_r)$. Concretely, by (35)–(36),

$$u^R(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ s = \frac{u_\ell + u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ s < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{46}$$

the last case corresponding to $\xi = 0$ lying inside the rarefaction fan of (36), i.e. $u^R(0) = (f')^{-1}(0) = 0$ for Burgers' flux.

### 8.4.3 Stability and convergence

**Proposition 8.4** (CFL condition)**.** *The Godunov scheme is stable provided the time step satisfies*

$$\Delta t \le \frac{\Delta x}{\max_j |f'(U_j^n)|}. \tag{47}$$

Condition (47) (the Courant–Friedrichs–Lewy condition) ensures that information does not propagate across more than one cell per time step, consistent with the finite propagation speed of characteristics established in Section 2.

**Theorem 8.5** (Convergence; see [2] and standard finite volume references)**.** *Under* (47)*, the Godunov approximations converge, as $\Delta x, \Delta t \to 0$, to the unique entropy solution of Theorem 6.4.*

*Remark* 8.6. This gives a second, independent route to the solutions constructed by hand in Sections 5–7: an implementation of (45) applied to the Riemann data (27) should reproduce the shock speed $s = 1/2$ and the linear profile $u = x/t$ inside the rarefaction fan, providing a computational check on the exact formulas derived analytically above.

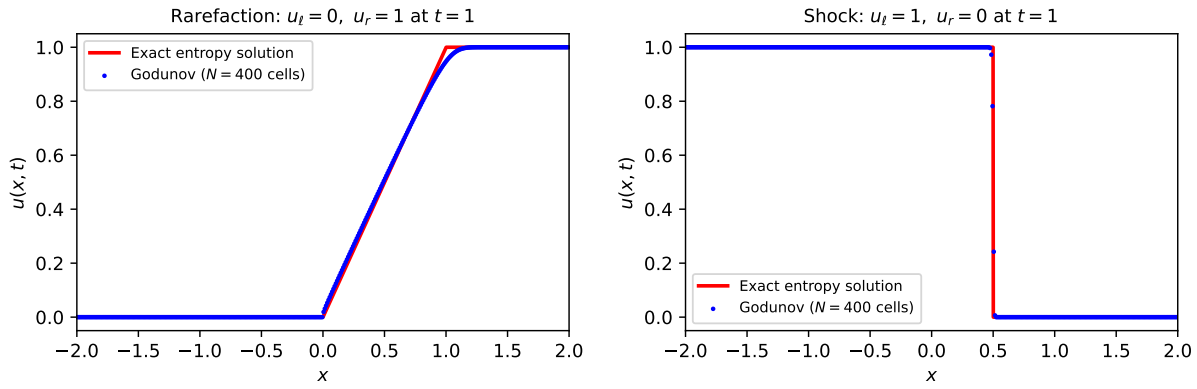



Figure 3: Godunov scheme (blue dots, $N = 400$ cells) against the exact entropy solution (red) of Theorem 7.1, at $t = 1$. Left: rarefaction data $(u_\ell, u_r) = (0, 1)$; the scheme reproduces the linear fan $u = x/t$, with mild smearing at the two corners typical of a first-order scheme. Right: shock data $(u_\ell, u_r) = (1, 0)$; the discontinuity is captured sharply at $x = st = 0.5$, exactly the Rankine–Hugoniot speed from (26), with no spurious oscillation.

### 8.4.4 Numerical results

The scheme of Section 8.4.1–8.4.2 was implemented with $N = 400$ cells on $x \in [-2, 2]$, using the CFL number 0.45 in (47), and run to $t = 1$ on both Riemann problems considered earlier: the rarefaction data $(u_\ell, u_r) = (0, 1)$ of Section 5, and the shock data $(u_\ell, u_r) = (1, 0)$. Figure 3 compares the numerical solution against the exact entropy solution of Theorem 7.1.

The measured $L^1$ errors between the numerical and exact solutions are approximately $4.5 \times 10^{-3}$ (rarefaction case) and $1.2 \times 10^{-3}$ (shock case), consistent with the first-order convergence rate of Theorem 8.5; the shock case converges faster per cell because the error is concentrated in the single cell straddling the discontinuity, whereas the rarefaction error accumulates gradually across the two fan corners.

## 8.5 Long-time behaviour: the N-wave

This final extension describes the universal shape that entropy solutions with compactly supported initial data approach as $t \to \infty$, regardless of the detailed shape of $u_0$.

### 8.5.1 An exact self-similar solution

**Proposition 8.7** (The N-wave)**.** *Fix constants $P, Q > 0$. The function*

$$N(x,t) = \begin{cases} 0, & x \le -\sqrt{2Qt}, \\ x/t, & -\sqrt{2Qt} < x < \sqrt{2Pt}, \\ 0, & x \ge \sqrt{2Pt}, \end{cases} \qquad t > 0, \tag{48}$$

*is an entropy solution of $u_t + u\,u_x = 0$ on $\mathbb{R} \times (0, \infty)$.*

*Proof.* On the open middle interval, $N(x,t) = x/t$ is exactly the self-similar rarefaction profile of Section 7.1, which solves the PDE classically there; the outer regions are trivially constant solutions. At the right edge $x_+(t) = \sqrt{2Pt}$, the one-sided limits are $u_\ell = x_+(t)/t = \sqrt{2P/t}$ and $u_r = 0$; since $\dot{x}_+(t) = \sqrt{P/(2t)} = \frac{1}{2}\big(\sqrt{2P/t} + 0\big)$, this is exactly the Rankine–Hugoniot speed (26), and $u_\ell > u_r$ so the Lax condition (32) holds. The same check at the left edge $x_-(t) = -\sqrt{2Qt}$, where $u_\ell = 0$ and $u_r = -\sqrt{2Q/t}$, gives $\dot{x}_-(t) = -\sqrt{Q/(2t)} = \frac{1}{2}\big(0 + (-\sqrt{2Q/t})\big)$, again matching (26), with $u_\ell = 0 > -\sqrt{2Q/t} = u_r$ satisfying the Lax condition. Hence both jumps are entropy-admissible, and $N$ is an entropy solution by Theorem 4.3 and Definition 6.1. □

The profile (48) is called an *N-wave*: it rises linearly through the origin and is bounded by a shock on each side, so that its graph traces the shape of the letter N (reversed), decaying in amplitude like $t^{-1/2}$ and spreading in width like $t^{1/2}$ as $t \to \infty$. It is an exact solution for every $t > 0$, though its formal limit as $t \to 0^+$ is not an ordinary function of the type considered in Section 4: both edges collapse to the origin, consistent with $N(\cdot, t)$ representing, in the limit, a state concentrated entirely at a point.

### 8.5.2 Shock merging

The convergence result below relies on one further fact: colliding shocks merge into one, governed by exactly the Riemann-problem machinery already developed.

**Lemma 8.8** (Shock merging)**.** *Let $u_1 > u_2 > u_3$, and consider initial data equal to $u_1$ for $x < a$, $u_2$ for $a < x < b$, and $u_3$ for $x > b$. For small $t > 0$ the entropy solution consists of two shocks moving at speeds $s_{12} = (u_1 + u_2)/2$ and $s_{23} = (u_2 + u_3)/2$; since $s_{12} > s_{23}$, they collide at the finite time $t^\times = 2(b - a)/(u_1 - u_3)$, after which the entropy solution is the single shock connecting $u_1$ to $u_3$ at speed $s_{13} = (u_1 + u_3)/2$.*

*Proof sketch.* Each jump individually satisfies the Lax condition since $u_1 > u_2 > u_3$, so Theorem 7.1(a) gives the two shock speeds above. Since $s_{12} - s_{23} = \frac{1}{2}(u_1 - u_3) > 0$, the shock positions $a + s_{12}t$ and $b + s_{23}t$ meet at the stated finite $t^\times$. At that instant the middle state has shrunk to zero width, leaving the two-state data $u_1, u_3$; since $u_1 > u_3$, Theorem 7.1(a) applied afresh at $(x^\times, t^\times)$ gives the merged shock at speed $s_{13}$. □

### 8.5.3 Convergence of general solutions to the N-wave

**Theorem 8.9** (Asymptotic N-wave decay; Lax, 1957 [4])**.** *Let $u_0 \in L^1(\mathbb{R})$ have compact support, and set*

$$P = \int_{\mathbb{R}} \max(u_0(x), 0)\, dx, \qquad Q = \int_{\mathbb{R}} \max(-u_0(x), 0)\, dx. \tag{49}$$

*Then the entropy solution $u$ of* (6) *satisfies*

$$\|u(\cdot, t) - N(\cdot, t)\|_{L^1(\mathbb{R})} \to 0 \qquad \text{as } t \to \infty, \tag{50}$$

*where $N$ is the N-wave* (48) *with the same constants $P, Q$.*

The proof is not reproduced here, but the mechanism is exactly Lemma 8.8: a compactly supported $u_0$ produces, for small $t$, a solution with possibly many shocks and rarefactions, but every pair of shocks that meet coalesces into one, so only finitely many discontinuities survive as $t \to \infty$. Since the total signed area $\int u(x, t)\, dx = P - Q$ is conserved for all $t$ (integrate (22) over all $x$), the surviving structure is forced into the unique self-similar shape consistent with these boundary areas — exactly (48).

# 9 Conclusion

This paper has given a complete account of Burgers' equation, from its origin as a simplified model of nonlinear advection and viscous diffusion (Section 1) to the theory required to solve it once classical solutions cease to exist. The method of characteristics (Section 2) gives an explicit solution formula that is exact but necessarily temporary: Section 3 showed that decreasing initial data produces a genuine, finite-time loss of smoothness, with an exact formula for the time and location of the first singularity. Making sense of the equation beyond this point required weakening the notion of solution (Section 4), which in turn revealed a genuine failure of uniqueness (Section 5): the same initial data can evolve into more than one weak solution. The Lax entropy condition (Section 6) was introduced precisely to resolve this, and Section 7 used it to give the complete, explicit entropy solution of the Riemann problem in both possible cases.

The extensions in Section 8 showed that this core theory extends beyond the fully worked examples above. The Cole–Hopf transformation justified the entropy condition itself, as the selection principle recovered in the vanishing-viscosity limit rather than an arbitrary rule, and showed that viscosity alone, without the vortex-stretching mechanism absent from this one-dimensional model, is enough to guarantee global smoothness. The Godunov scheme, built directly on the exact Riemann solver derived here, reproduced the analytic shock speed and rarefaction profile numerically to within first-order error. Finally, the N-wave result showed that the long-time behaviour of any compactly supported solution is governed by a single universal profile, determined only by the total positive and negative area of the initial data, via the same shock-merging mechanism used to build the Riemann solver itself.

Taken together, these results illustrate why Burgers' equation, despite its simplicity, remains a standard reference point in the theory of nonlinear partial differential equations: it is simple enough to solve exactly and completely, in every regime considered here, while already exhibiting the central phenomena — shock formation, non-uniqueness of weak solutions, the necessity of an entropy condition, and the regularising effect of viscosity — that make the general theory of nonlinear conservation laws, and by extension the far harder open questions surrounding the Navier–Stokes and Euler equations, so difficult to resolve.

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